Efficient Coding of Drifting Images or maybe Statistics of Moving Images

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Abstract

Abstract here

1 Introduction

- Eyes are never still. Saccades allow for looking at things without moving head. But even when head is still and eyes are fixating, movements persist.
- Such fixational movements have been shown to be a feature, not a bug.
-
- Efficient coding principles provide insight into function and design of early sensory systems

Goal is to understand the statistical structure of images formed on moving sensors (such as animal eyes) and understand it's implications on retinal coding. Deriving a complete statistical description of images formed in moving optical systems is difficult. We make the simplifying assumption that the statistics of the movement changes in stationary increments and independent of the scene being imaged and analyze pairwise statistics. Under these assumptions, we can derive an analytical form of the autocorrelation function $C_z(\xi, \tau)$, which can then be studied in the Fourier domain. The power spectrum is generally not separable in spatial and temporal frequencies. We study the structure of the power spectrum arising from power-law natural images undergoing Brownian motion. Assuming well-known statistical structure of natural images, the resulting power spectrum has interesting properties. Specifically, slicing at a spatial or

temporal frequency, power is flat up to a cut-off frequency, and then falls as a function of f^{-4} or ω^{-2}. Derive efficient coding models that predict optimal filters that are center-surround.

Statistical structure of drifting natural images (either under empirically observed drifting eye motion or Brownian) has been reported before (Rucci ....). Here, we seek describe the statistical structure of the data, and understand its implications for retinal coding.

2 Notes on notation

Fourier transform of space x and time t functions is indicated by their arguments k, ω. I'm going to assume space is one dimensional for derivation throughout as it keeps notation simple. The extension to 2d space is straightforward.

3 Analysis

3.1 Statistical Structure of Moving Image Sensors

Consider writing the signal as $I(x,t) = I_0(x - u(t))$.

Assume a static image $s(x)$ where x denotes position. The time-varying process $u(t)$ changes the position as a function of time,

$$z(x,t) = s\big(x - u(t)\big), \tag{1}$$

producing images $z(x,t)$ that can vary in both space and time. To make this problem analytically tractable, we will assume that the the movement $u(t)$ is stochastic, stationary in time t, and independent of static signal $s(x)$. Under the assumptions of $u(t)$ above, the probability density of displacements is a function of spacial displacement ξ and temporal lag τ, given by

$$u(t+\tau) - u(t) \sim \rho(\xi, \tau). \tag{2}$$

The space-time autocorrelation $C_z(\xi,\tau)$ can be written as a simple spatial convolution of the spacial autocorrelation function $C_s(\xi)$ and the movement density function $\rho(\xi,\tau)$,

$$C_z(\xi,\tau) = \big\langle z(x,t)\, z(x+\xi, t+\tau)\big\rangle_{x,t} \tag{3}$$

$$= \big(\rho(\cdot,\tau) * C_s\big)(\xi) \tag{4}$$

See appendix A.1 for derivation details. Given the spatial convolution above and the image autocovariance function's independence of time, the space-time power spectrum $C_z(k,\omega)$ of z reduces to simple multiplication in Fourier domain

$$C_z(k,\omega) = \rho(k,\omega)\, C_s(k). \tag{5}$$

Points to make

- Spatiotemporal power spectrum is not generally not separable. Contrast to Ocko, Jun, etc.
- Movement is a filter on the image that pushes spatial information into time.
- If image is also changing in time (i.e., $s(x,t)$), you still get convolution over space only, but its autocorrelation is now $C_s(\xi,\tau)$. This becomes a bit tricky in the Fourier domain because now it isn't simple multiplication.

3.2 Drift

Mathematical details of Brownian motion (i.e., the process by which brownian motion is generated and the resulting density function $\rho(\xi,\tau)$).

motivate drift. eye is never still. well approximated by drift. Drift is a feature not a bug. Exploited by imaging technology: google(?) paper that uses motion from hand tremor to do subpixel resolution. [likely move this to introduction]

Let $u(t)$ evolve stochastically from initial position x_0 via Brownian motion W_t with diffusion constant D:

$$du(t) = \sqrt{2D}dW(t), \quad u(0) = x_0. \tag{6}$$

The probability distribution of the SDE solution is $\rho(\xi,\tau)$, where ξ and τ is space and time, respectively

$$\rho(\xi,\tau) = \frac{1}{\sqrt{4\pi D|\tau|}} \exp\left(-\frac{\xi^2}{4D|\tau|}\right). \tag{7}$$

The space-time fourier transform of $\rho(\xi,\tau)$ has a simple form (see appendix A.2 for derivation details). A well known property of static natual images is that the power spectrum is power-law in spatial frequency. This means that $C_s(k) \propto k^{-\beta}$ for some $\beta > 0$. The resulting space-time power spectrum is then given by the power spectrum of the static signal $C_s(k)$ multiplied by a simple term derived from the spate-time Fourier transform of ρ

$$C_z(k,\omega) = \rho(k,\omega)\, C_s(k) \tag{8}$$

$$= \frac{2Dk^{2-\beta}}{(Dk^2)^2 + \omega^2} \tag{9}$$

for $D > 0$. The power spectrum for brownian motion drifted images takes a a simple form.

3.3 Information maximizing linear filters

Not convinced that the spatiotemporal descriptions are necessary. Can just start with Fourier domain. Then the power constraint is more intuitive as well...

To ask how drift shapes efficient retinal coding, we model the response using a linear space-time filter w. The response $r(x,t)$ of the linear model is a function of the input $z(x,t)$ and noise $n_{in}(x,t)$ and output noise $n_{out}(x,t)$,

$$y(x,t) = \big(w * (z + n_{in})\big)(x,t), \quad r(x,t) = y(x,t) + n_{out}(x,t). \tag{10}$$

Here, $*$ denotes convolution over space and time. We take the input and output noise to be independnt Gaussian noises with zero mean, with variance σ_{in}^2 and σ_{out}^2, respectively. We seek to find the filter w and drift constant D that maximizes the mutual information between the input z and response r, subject to a power and temporal derivative constraint on the response.

Constraints on the filter are necessary because, without them, the gain of the linear filter can be increased arbitrarily. Output power is constrained (similar to Atick and Redlich). We introduce an additional penalty on the temporal derivative of the outputs as high frequency responses to not come for free in neural systems, they require larger neurons and increased metabolic cost. We constrain the power of the filtered signal before output noise is added. The constrained optimization problem is as follows

$$\begin{aligned} \underset{w,D}{\text{maximize}} \quad & I(Z;R) \\ \text{subject to} \quad & \left\langle y(x,t)^2 \right\rangle_{x,t} + \eta \left\langle \dot{y}(x,t)^2 \right\rangle_{x,t} = 1. \end{aligned} \tag{11}$$

where $\eta \geq 0$ controls the strength of the temporal smoothness penalty. The procedure for solving this optimization problem is described in Appendix A.3.

4 Results

4.1 Brownian drift of natural images

It is well known that the power spectrum of natual iamges have is power-law in spatial frequency with an exponent of about 2. When $\beta = 2$, the spatial frequency term in equation 9 in the numerator is canceled. This has a whitening effect on the power spectrum (figure 1). This qualitative effect has been described previously. Here, we've developed an analytic model that capture this effect. The whitening effect is seen more clearly by looking at cross sections at various spatial frequencies and temporal frequencies (figure 1). The power spectrum is flat up to a cut-off frequency, and then falls as a function of f^{-4} or ω^{-2}. This cut off frequency is at $\omega = Dk^2$.

The fall of the power spectrum is quite different from previous reports of the statistics of natural movies (Dong and Atick, Jun et al., ...). Dong and Atick analyzed Handycam footage and scenes from indiana jones and found a non-seperable power spectrum that falls that has an exponent between $-1, -2$ for both spacial and temporal frequencies. World-fixed camera footage of natural scenes are reported to have a space-time seperable power spectrum that falls as $C(k,\omega) \propto k^{-2}\omega^{-1.3}$ (Jun et al.). Ocko also considers a space-time seperable power spectrum.

These charactizations of space-time statistics of natural image indicate that they contain correlations in both space and time. Decorrelation or whitening of correlated

sensory inputs is a classic theory of retinal processing (atick and redlich, olshausen and field, Doi, etc.) that predicts circularly symmetric center-surround receptive fields as whitening filters.

Drifting eye movements has an apparent whitening effect on natural images (figure 1) as the power spectrum is flat up to a cut-off frequency. This raises a conundrum. If the input spectrum to the retina has already been whitened by eyemovements, what are center surround receptive fields doing in the retina? Figure 1 notes

- Describe how the power spectrum of drifting images depends on the static-image exponent β and diffusion constant D. Increasing drift spreads power toward higher temporal frequencies, converting spatial structure into temporal fluctuations.
- Changing β changes the spatial slopes. The natural-image value $\beta = 2$ is special because it cancels the spatial-frequency dependence in the numerator, producing a flat spatial spectrum below the knee at fixed nonzero temporal frequency.
- For $\beta = 2$, spatial and temporal cross sections are each approximately flat below their respective knees. The analytical crossover is $|\omega| = Dk^2$: the spatial knee is $k = \sqrt{|\omega|/D}$ and the temporal knee is $|\omega| = Dk^2$.
- The two cross sections have different high-frequency fall-offs: k^{-4} in space and ω^{-2} in time.
- This partial whitening raises the central question: if drift has already whitened the retinal input over part of the spectrum, what role remains for center-surround receptive fields? Address this using information-maximizing linear filters.

4.2 Optimal linear filters are spatially bandpass

Figure 2 notes

- Use a reference set of noise parameters consistent with the retina. Jointly optimizing the filters and diffusion constant without a temporal constraint favors extremely large diffusion constants. Introduce a temporal smoothness penalty to constrain this behavior. choose the temporal penalty strength, then hold that strength fixed.
- With the temporal constraint, mutual information as a function of diffusion has an optimum for a given set of noise parameters.
- The noise dependence of optimal diffusion is asymmetric: increasing input noise increases the optimal diffusion constant, whereas increasing output noise decreases it.

Figure 3 notes

- Examine the optimal filters at a fixed diffusion constant of $D = 40$ arcmin^2/s and fixed input and output noise levels. Compare the signal spectrum with the optimal spatiotemporal filter.

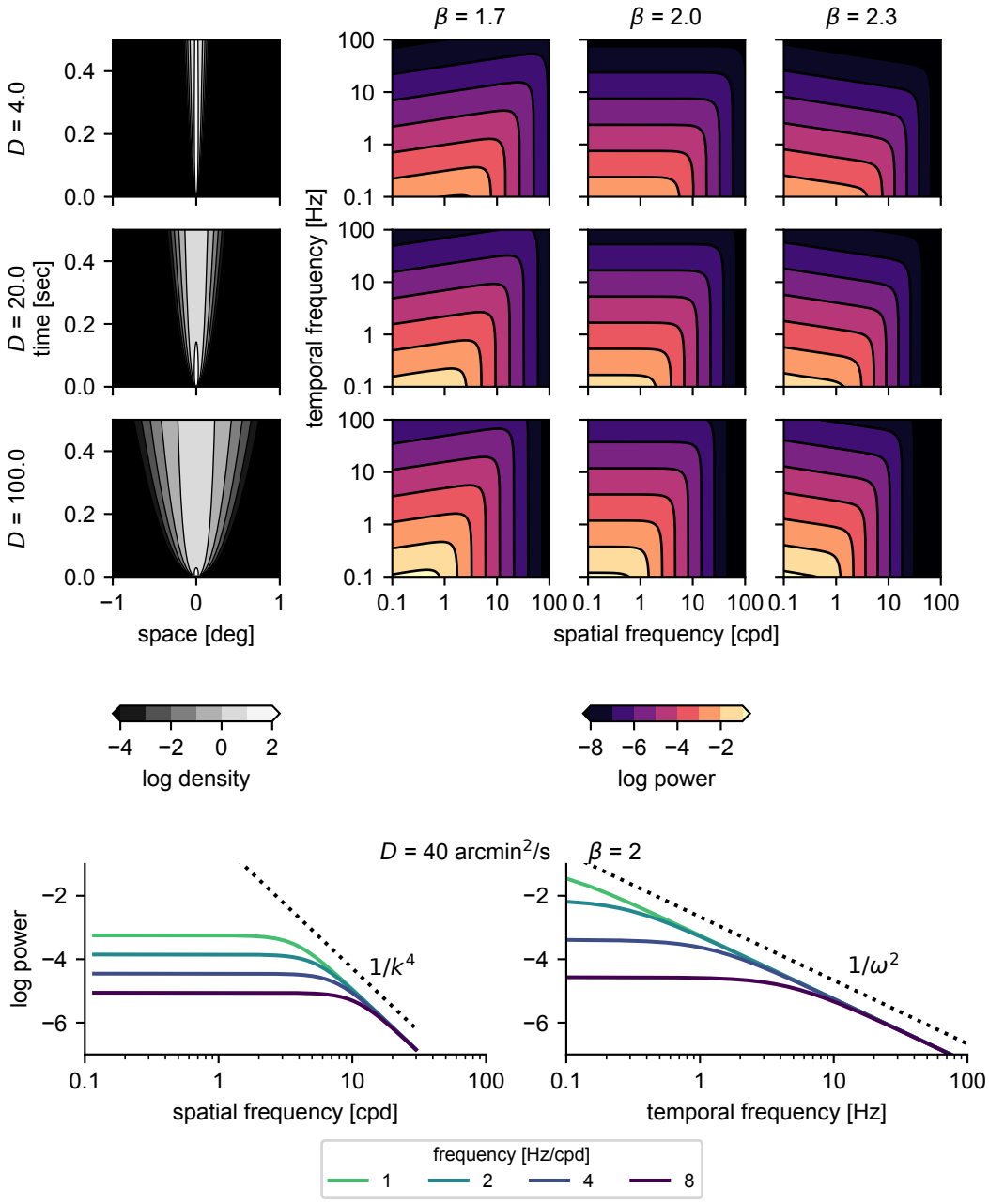


Figure 1: spatio temporal power spectra for different betas. Diffusion values are in arcmin^2/s.

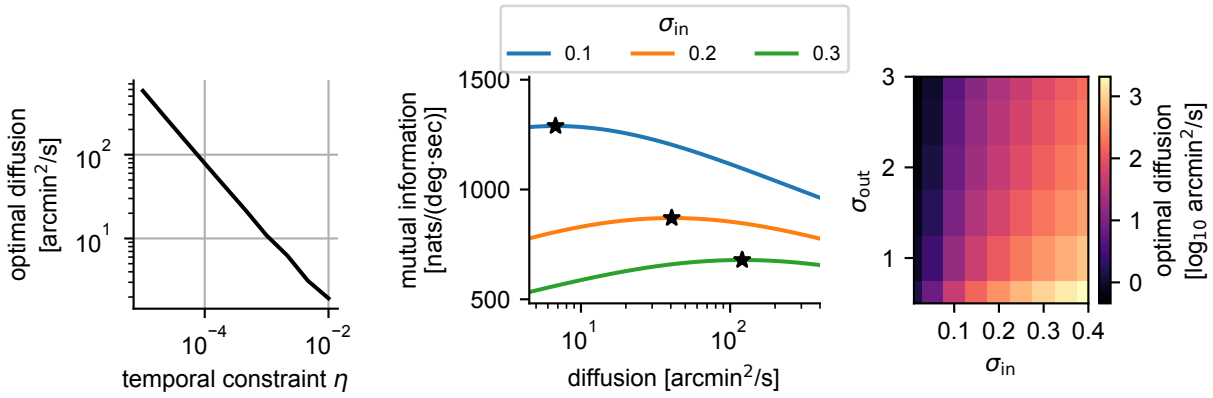


Figure 2: Optimal diffusion constant vs. input and output noise.

- The optimal filter is spatially bandpass. It amplifies spatial frequencies beyond the knee, where signal power falls as k^{-4}, up to the range limited by the input noise floor.
- Convert the optimal frequency-domain filter into spatial and temporal kernels using the phase choices described in the appendix. The spatial kernel has a center-surround structure, and the temporal frequency response is approximately low pass.
- This resolves the a conundrum: drift whitens only part of the spectrum. Amplifying the remaining spatial fall-off still favors center-surround filtering under the efficient coding objective.

Figure 4 notes

- Compare how the optimal kernels change with input noise when diffusion is allowed to adapt versus when it is held fixed.
- With adaptive diffusion, increasing input noise increases the optimal diffusion constant. The kernels become broader in space and slightly faster in time while retaining their qualitative center-surround structure.
- With fixed diffusion, increasing input noise weakens the surround and makes the optimal filter more spatially low pass, consistent with the Atick and Redlich result and the empirical trend described in these notes.
- Allowing drift to adapt therefore changes the predicted response to input noise: spatial broadening with maintained surrounds, compared with loss of surrounds when drift is fixed.

other notes

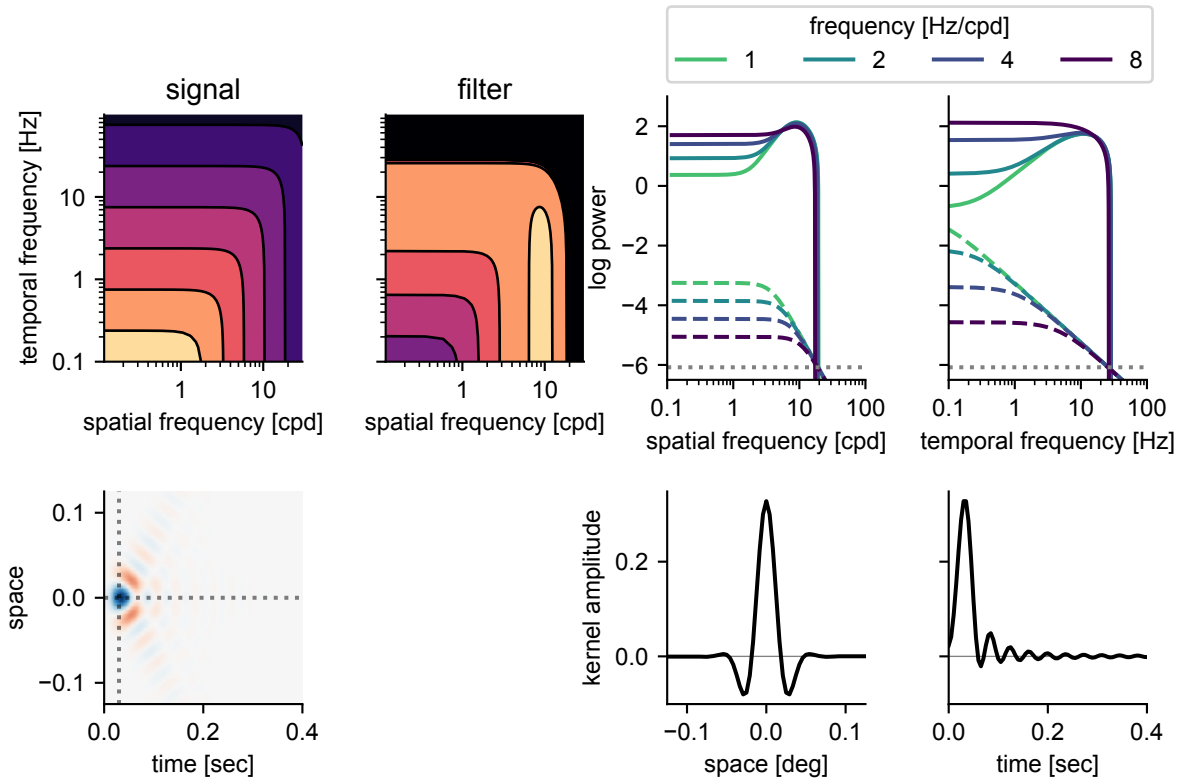


Figure 3: Optimal filter and kernel.

- https://www.jneurosci.org/content/jneuro/34/38/12701.full.pdfm – aytekin victor rucci 2014 – The results of our study show that, during normal fixation, the spatiotemporal signals impinging on retinal receptors are already whitened within the range of temporal frequencies at which ganglion cells respond best. Whats our response? This whitening occurs before any neural processing takes place. "Thus, neural decorrelation, is not a viable theory to explain the filtering characteristics of receptive fields. contrast sensitivity functions of retinal ganglion cells further enhance high spatial frequencies above and beyond what is required for redundancy removal, leading to an enhancement of the luminance discontinuities in the scene. Critically, this edge enhancement happens in the temporal domain: it occurs in the synchronous firing of retinal ganglion cells, yielding a temporal code that takes advantage of the notion that synchronous responses tend to propagate more reliably than asynchronous ones."

5 Optimal weights for linear-nonlinear model

Linear model is limited as it only considers pair-wise correlations as relies on power spectrum which is equal to fourier transform of autocorrelation function. Its well known that natural images contain higher-order correlations. A natural extension of the linear model (descirbed by Atick and Redlich, Linsker) is the linear-nonlinear model of karklin and simoncelli, further extended to space-time by Jun et al. Linsker showed that

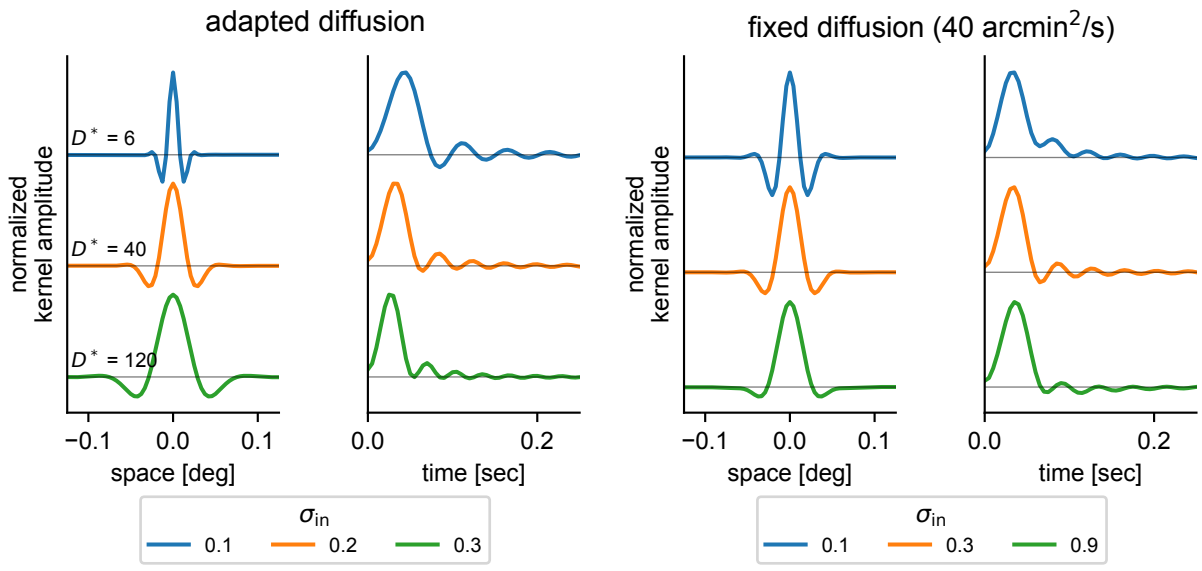


Figure 4: Optimal diffusion constant vs. input and output noise.

introducting nonlinearities into the model allows it to learn higher-order correlations. While K&S motivation to introduce nonlinearities is to model nonnegative code in the optic nerve (biological mimicry), nonlinearities also have the pricipled motivation of allowing the model to learn higher-order correlations.

Describe the model

Results:

- Weights become spatially localized.
- They bifurcate into three classes. Each class bifucates into on and off types. each subtype 1 and 2 tiles space. 3 hasn't gotten there yet.
- type 1 and 2 spatial filters are bandpass, consistent with retina and linear model.

Discussion

- Main contributions: Describing pair-wise statistical structure of stochastically moving images. Focus in on drift, which produces a whitening effect. Despite this apparent structure, we show that circularly symmetric center-surround receptive fields of RGCs are predicted (optimal) under efficient coding objective. Gained understanding in linear model; nonlinear model is consistent. The reconciliation relies on the specific structure of the power spectrum. spatial Power drops as k^{-4} after a cut-off frequency. Still signal left to amplify, thus optimal filters become band-pass in space.
- Generate new prediction: higher noise (low contrast) eyes may drift more? Drifting allows for reshaping statistics before any neural processing occurs.

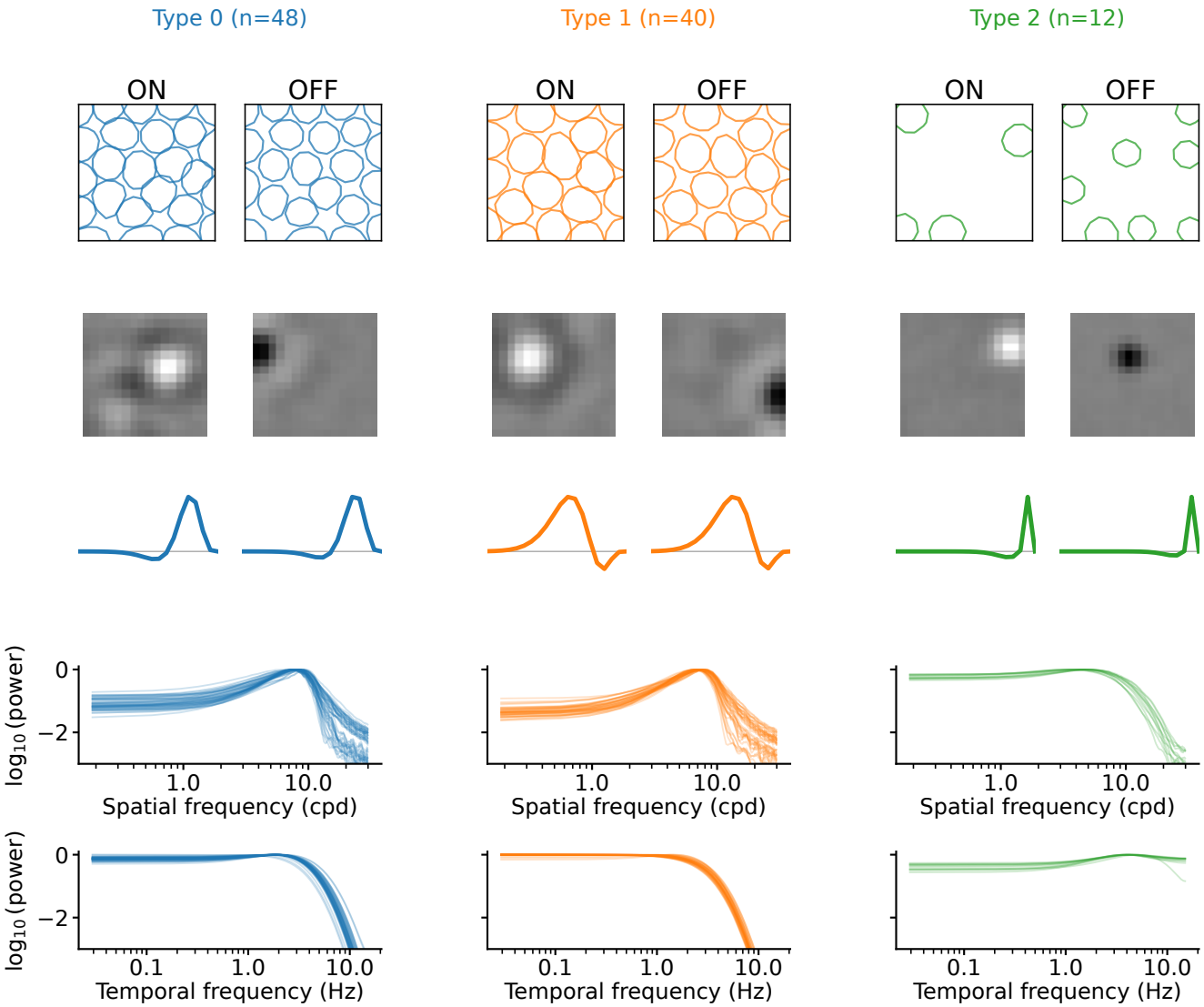


Figure 5: Optimal weights for linear nonlinear model.

- Relevant work – rucci.
- Earlier reports of spatio temporal statistics of natural images – Dong and Atick, Eckea M P and Buchsbaum G 1993,
- Efficient coding literature – Jun, Ocko.
- Nonseparable power spectrum. Receptive fields reported to be separable in space time. Maybe due to receptive field measurement procedure. Nonseperable receptive fields could be produced from temporal dynamics of center and surround mechanisms in retina.
- Nonstationarity – saccades.

A Appendix

A.1 Autocorrelation of stochastically moving images

Follow logic of Dong and Attick 1995 (Dong and Atick, 1995). That is, get an equation that describes the spatio-temporal correlations $C_z(\xi, \tau)$ as a function of space ξ and time τ lags. Then we can take the fourier transform in space and time. Define spatial-temporal autocorrelation function

$$C_z(\xi, \tau) = \mathbb{E}_{x,t}\Big[z(x,t)\, z(x+\xi, t+\tau)\Big] \tag{12}$$

$$= \mathbb{E}_{x,t}\Big[s\big(x - u(t)\big)\, s\big(x + \xi - u(t+\tau)\big)\Big]. \tag{13}$$

We assume that $u(t)$ is governed by a stochastic process. For a temporal lag τ, define the spatial displacement

$$\Delta u(\tau) := u(t+\tau) - u(t), \tag{14}$$

where the density of this displacement is given by $\Delta u(\tau) \sim \rho(\xi, \tau)$ and ξ denotes a possible displacement value. Let $l = x - u(t)$ and substitute the stimulus $z(x,t)$ above with $s(x - u(t))$:

$$C_z(\xi, \tau) = \mathbb{E}_{x,t}\Big[s\big(x - u(t)\big)\, s\big(x + \xi - u(t+\tau)\big)\Big] \tag{15}$$

$$= \mathbb{E}_{l,t}\Big[s(l)\, s\big(l + \xi - \Delta u(\tau)\big)\Big] \tag{16}$$

Define the static image autocorrelation

$$C_s(\xi) := \mathbb{E}_x[s(x)s(x+\xi)]. \tag{17}$$

Assuming translation invariance, we can take expectation over brownian paths which then describes the spatiotemporal correlation, which can be equivalently described as spatial convolution

$$C_z(\xi, \tau) = \mathbb{E}_{\Delta u(\tau)}\Big[C_s\big(\xi - \Delta u(\tau)\big)\Big] \tag{18}$$

$$= \int_{-\infty}^{\infty} C_s(\xi - \delta)\, \rho(\delta, \tau)\, d\delta \tag{19}$$

$$= (C_s * \rho(\cdot, \tau))(\xi). \tag{20}$$

A.2 Spatio-temporal power spectrum of images undergoing Brownian translations

$$\rho(\xi, \tau) = \frac{1}{\sqrt{4\pi D|\tau|}} \exp\left(-\frac{\xi^2}{4D|\tau|}\right). \tag{21}$$

The spatial Fourier transform of Gaussian ρ is another Gaussian:

$$\rho(k, \tau) = e^{-D|\tau|k^2}. \tag{22}$$

Putting these two together, you get

$$C_z(k,\tau) = C_s(k)\, e^{-D|\tau|k^2} \tag{23}$$

Fourier transform in time Now take the Fourier transform in time τ:

$$C_z(k,\omega) = \int_{-\infty}^{\infty} C_z(k,\tau) e^{-i\omega\tau}\, d\tau \tag{24}$$

$$= C_s(k) \int_{-\infty}^{\infty} e^{-Dk^2|\tau|} e^{-i\omega\tau}\, d\tau. \tag{25}$$

The integral can be solved by splitting into two integrals from $(-\infty, 0)$ and $[0, \infty)$ to deal with the $|t|$ in exponent which ends up working out into a simple form

$$\int_{-\infty}^{\infty} e^{-a|t|} e^{-i\omega t}\, dt = \frac{2a}{a^2 + \omega^2} \tag{26}$$

with $a = Dk^2$, we get

$$C_z(k,\omega) = C_s(k)\, \frac{2Dk^2}{(Dk^2)^2 + \omega^2} \tag{27}$$

up to a constant. Note that in the limit as D goes to zero, all power lies at $\omega = 0$, and within this plane the power is the input static image power

$$\lim_{D \to 0^+} C_z(k,\omega) \propto C_s(k)\, \delta(\omega) \tag{28}$$

A.3 Optimal linear filters

The squared magnitudes of the optimal linear filters $v(k,\omega) = |w(k,\omega)|^2$ can be solved for analytically in the Fourier domain. The spatiotemporal response of the linear model is given by

$$r(k,\omega) = w(k,\omega)\big(z(k,\omega) + n_{in}(k,\omega)\big) + n_{out}(k,\omega). \tag{29}$$

We distinguish the space-time noise variances σ_{in}^2 and σ_{out}^2 from their power spectral densities. For Gaussain white noise, these densities are constant over the modeled frequency band and we denote their levels by N_{in} and N_{out}. Under the Gaussian input approximation, the mutual information rate per unit space between the input z and response r is

$$I(Z;R) = \frac{1}{2} \int \frac{dk, d\omega}{(2\pi)^2}\, \log\left(1 + \frac{v(k,\omega), C_z(k,\omega)}{N_{in} v(k,\omega) + N_{out}}\right). \tag{30}$$

The term in the integral has the familiar form of $\log\big(1 + \text{SNR}(k,\omega)\big)$, which is then integrated over spatial and temporal frequencies.

The combined power and temporal derivative constraint can also be written in terms of spatial and temporal frequencies as a function power spectrum of the filter response $C_y(k,\omega)$,

$$\int \frac{dk, d\omega}{(2\pi)^2} \left(1 + \eta\omega^2\right) C_y(k,\omega) = 1, \quad C_y(k,\omega) = v(k,\omega)\left(C_z(k,\omega) + N_{in}\right). \tag{31}$$

where ω^2 arises from the temporal derivative constraint.

The information optimization objective subject to this constraint can be rewritten using a Lagrange multiplier λ. The Lagrangian for a given spatial and temporal frequency is

$$\mathcal{L}_{k,\omega} = \frac{1}{2}\log\left(1 + \frac{v(k,\omega)C_z(k,\omega)}{N_{in}v(k,\omega) + N_{out}}\right) - \lambda(1+\eta\omega^2)v(k,\omega)\big(C_z(k,\omega) + N_{in}\big). \tag{32}$$

Setting $\partial\mathcal{L}k,\omega/\partial v = 0$ at an active mode gives

$$\begin{aligned}\frac{N_{out}C_z(k,\omega)}{2\big(N_{out} + (C_z(k,\omega)+N_{in})v(k,\omega)\big)\big(N_{out}+N_{in}v(k,\omega)\big)} \\ = \lambda(1+\eta\omega^2)\big(C_z(k,\omega)+N_{in}\big).\end{aligned} \tag{33}$$

For positive input and output noise levels, solving for $v(k,\omega) = |w(k,\omega)|^2$ yields

$$v(k,\omega) = \frac{N_{out}}{N_{in}}\left[\frac{1}{2}\frac{C_z(k,\omega)}{C_z(k,\omega)+N_{in}}\left(\sqrt{1 + \frac{N_{in}}{N_{out}}\frac{2}{\lambda(1+\eta\omega^2)C_z(k,\omega)}} + 1\right) - 1\right]_+. \tag{34}$$

The nonlinearity $[\cdot]_+ = \max(\cdot, 0)$ ensures that filter powers $v(k,\omega) = |w(k,\omega)|^2$ are nonnegative. The Lagrange multiplier λ that satisfies the constraint is found by starting from an initial λ, computing $v(k,\omega)$ via the equation above, and then adjusting λ up or down depending on whether the cost exceeds or undershoots the budget.

A.4 Conversion to space-time

We reconstruct the space–time kernel $w(x,t)$ by inverse Fourier transforming the optimal filter $w(k,\omega) = \sqrt{v(k,\omega)}\, e^{i\theta(k,\omega)}$:

$$w(x,t) = \int \frac{dk, d\omega}{(2\pi)^2}\, w(k,\omega)\, e^{i(kx+\omega t)}.$$

The optimization determines the filter amplitude but leaves its phase $\theta(k,\omega)$ unspecified. We choose a spatially symmetric, emporally minimum-phase response. Since $v(k,\omega)$ is even in k, spatial symmetry is preserved by assigning the same phase to positive and negative spatial frequencies. The minimum temporal phase is given by the negative Hilbert transform of the log amplitude $\theta(k,\omega) = -\mathcal{H}_\omega\left\{\log\sqrt{v(k,\omega)}\right\}$.

References

Dong, D. W. and Atick, J. J. (1995). Statistics of natural time-varying images. *Network: Computation in Neural Systems*, 6(3):345–358.